Proportional hazards model

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Survival Analysis

Proportional hazards model

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```
load("Ahda_RLab3.RData")
library(survival)
```

A data set on patients with multiple myeloma ————————-

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```
str(multmyel)
```

```
'data.frame':	48 obs. of  9 variables:
 $ time  : int  13 52 5 40 10 7 66 10 10 14 ...
  ..- attr(*, "label")= chr "Survival time [from diagnosis]"
  ..- attr(*, "units")= chr "Month"
 $ status: int  1 0 1 1 1 0 1 0 1 1 ...
  ..- attr(*, "label")= chr "Death indicator"
 $ age   : int  66 66 53 69 65 57 52 60 70 70 ...
  ..- attr(*, "label")= chr "Age"
 $ sex   : Factor w/ 2 levels "Female","Male": 2 2 1 2 2 1 2 2 2 2 ...
  ..- attr(*, "label")= chr "Gender"
 $ BUN   : int  25 13 15 10 20 12 21 47 37 40 ...
  ..- attr(*, "label")= chr "Blood urea nitrogen"
 $ CA    : int  10 11 13 10 10 8 10 9 12 11 ...
  ..- attr(*, "label")= chr "Serum calcium"
 $ HB    : num  14.6 12 11.4 10.2 13.2 9.9 12.8 14 7.5 10.6 ...
  ..- attr(*, "label")= chr "Hemoglobin"
 $ PC    : int  18 100 33 30 66 45 11 70 47 27 ...
  ..- attr(*, "label")= chr "Percentage of white blood cells"
 $ BJP   : Factor w/ 2 levels "Yes","No": 1 2 1 1 2 2 1 1 2 2 ...
  ..- attr(*, "label")= chr "Bence-Jones protein"
 - attr(*, "comment")= chr "Survival times of patients with multiple myeloma"
```

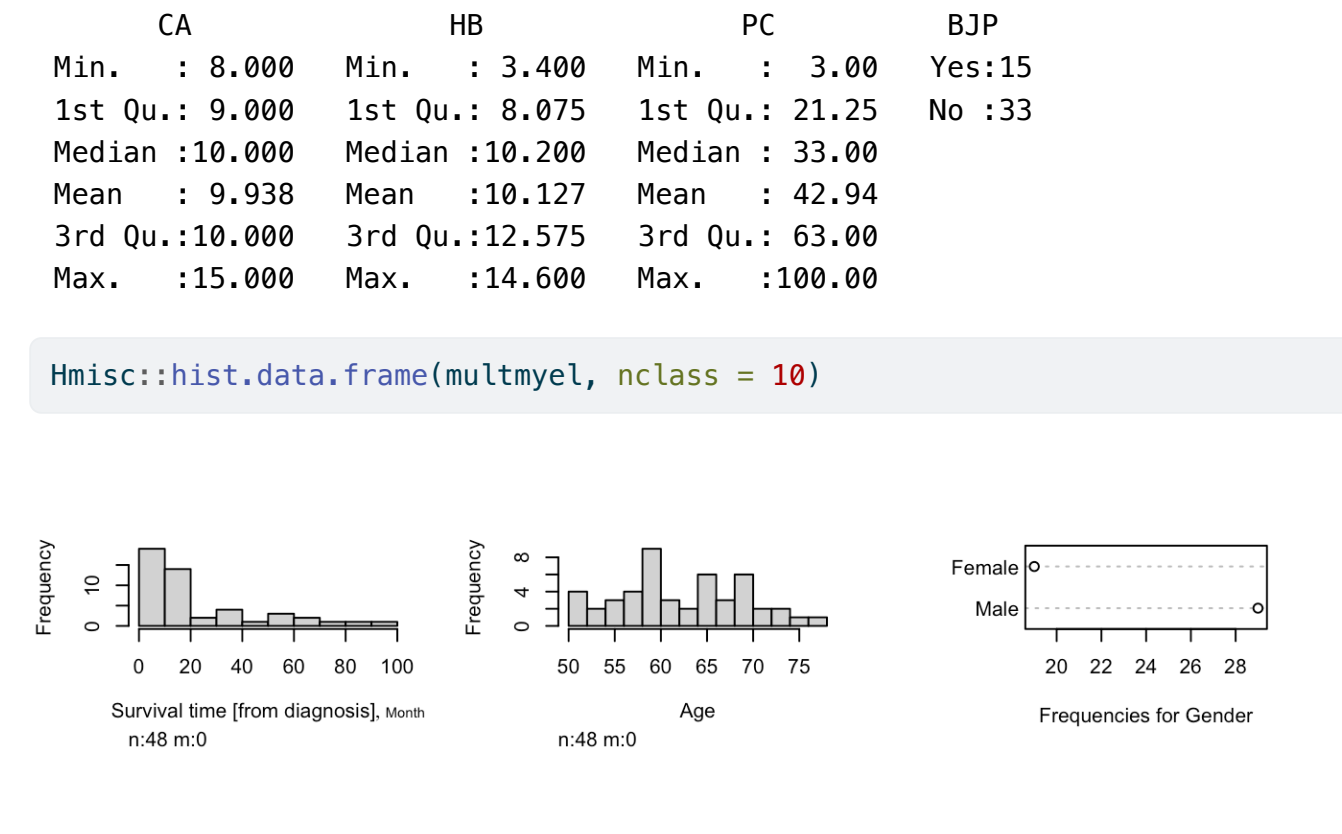
```
head(multmyel)
```

```
  time status age    sex BUN CA   HB  PC BJP
1   13      1  66   Male  25 10 14.6  18 Yes
2   52      0  66   Male  13 11 12.0 100  No
3    5      1  53 Female  15 13 11.4  33 Yes
4   40      1  69   Male  10 10 10.2  30 Yes
5   10      1  65   Male  20 10 13.2  66  No
6    7      0  57 Female  12  8  9.9  45  No
```

```
summary(multmyel)
```

```
      time           status          age             sex          BUN        
 Min.   : 1.00   Min.   :0.00   Min.   :50.00   Female:19   Min.   :  6.00  
 1st Qu.: 6.75   1st Qu.:0.75   1st Qu.:57.75   Male  :29   1st Qu.: 13.75  
 Median :14.50   Median :1.00   Median :61.50               Median : 21.00  
 Mean   :23.33   Mean   :0.75   Mean   :62.69               Mean   : 34.04  
 3rd Qu.:37.00   3rd Qu.:1.00   3rd Qu.:68.25               3rd Qu.: 39.25  
 Max.   :91.00   Max.   :1.00   Max.   :77.00               Max.   :172.00  
       CA               HB               PC          BJP    
 Min.   : 8.000   Min.   : 3.400   Min.   :  3.00   Yes:15  
 1st Qu.: 9.000   1st Qu.: 8.075   1st Qu.: 21.25   No :33  
 Median :10.000   Median :10.200   Median : 33.00           
 Mean   : 9.938   Mean   :10.127   Mean   : 42.94           
 3rd Qu.:10.000   3rd Qu.:12.575   3rd Qu.: 63.00           
 Max.   :15.000   Max.   :14.600   Max.   :100.00           
```

```
Hmisc::hist.data.frame(multmyel, nclass = 10)
```



The Surv object

———————

```
myesurv <- with(multmyel, Surv(time, status))
```

Two survival plots (drawn with function survplot[rms])

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```
library(rms)
```

```
Loading required package: Hmisc
```

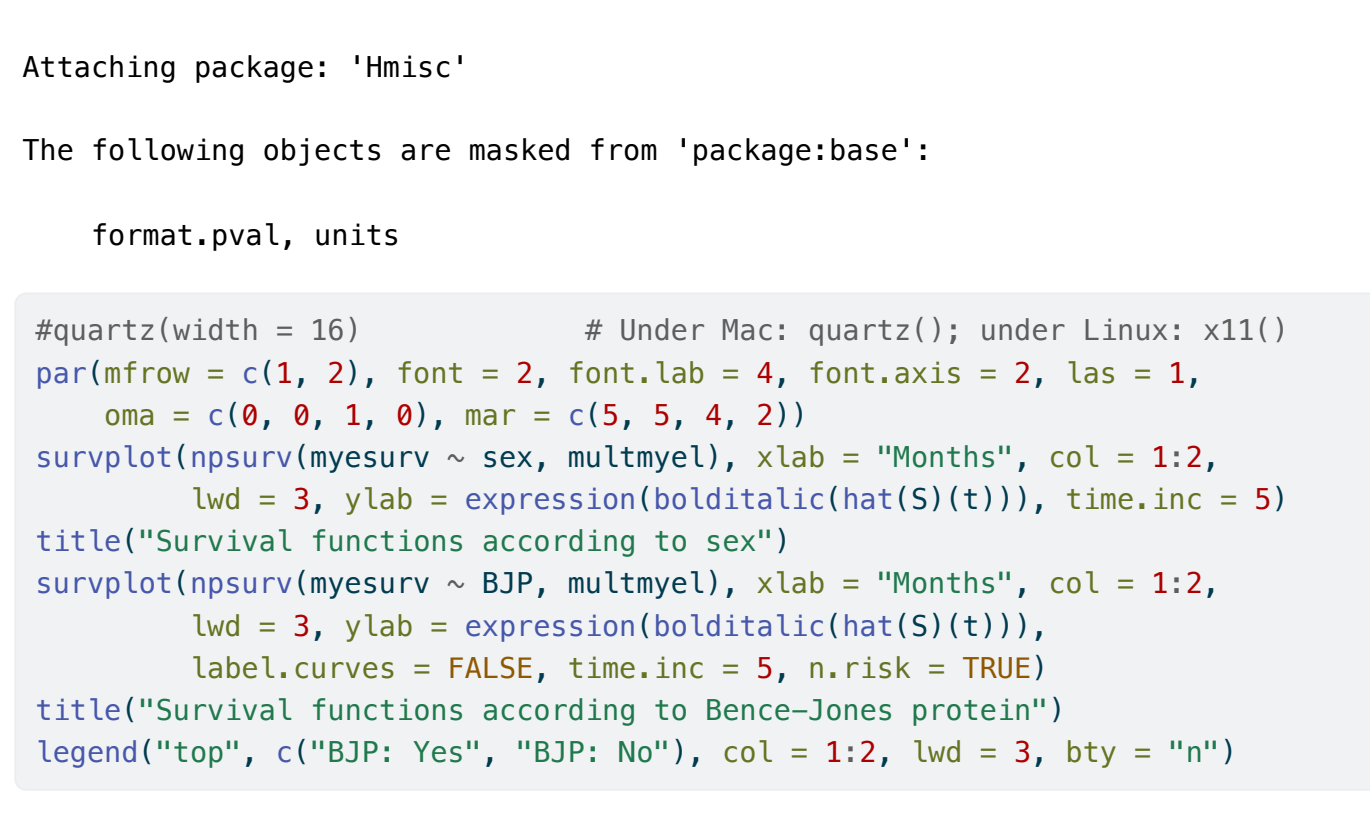
```

Attaching package: 'Hmisc'
```

```
The following objects are masked from 'package:base':

    format.pval, units
```

```
#quartz(width = 16)             # Under Mac: quartz(); under Linux: x11()
par(mfrow = c(1, 2), font = 2, font.lab = 4, font.axis = 2, las = 1,
    oma = c(0, 0, 1, 0), mar = c(5, 5, 4, 2))
survplot(npsurv(myesurv ~ sex, multmyel), xlab = "Months", col = 1:2,
         lwd = 3, ylab = expression(bolditalic(hat(S)(t))), time.inc = 5)
title("Survival functions according to sex")
survplot(npsurv(myesurv ~ BJP, multmyel), xlab = "Months", col = 1:2,
         lwd = 3, ylab = expression(bolditalic(hat(S)(t))),
         label.curves = FALSE, time.inc = 5, n.risk = TRUE)
title("Survival functions according to Bence-Jones protein")
legend("top", c("BJP: Yes", "BJP: No"), col = 1:2, lwd = 3, bty = "n")
```



Fit of Cox models—– ——————————————————

additative - needed for linearity in log!

BJP = protein (biomarker) ## ——————————————————————-

```
cox1 <- coxph(myesurv ~ sex + BJP, multmyel)
summary(cox1)
```

```
Call:
coxph(formula = myesurv ~ sex + BJP, data = multmyel)

  n= 48, number of events= 36 

            coef exp(coef) se(coef)      z Pr(>|z|)
sexMale -0.05852   0.94316  0.35707 -0.164    0.870
BJPNo    0.54382   1.72257  0.38863  1.399    0.162

        exp(coef) exp(-coef) lower .95 upper .95
sexMale    0.9432     1.0603    0.4684     1.899
BJPNo      1.7226     0.5805    0.8042     3.690

Concordance= 0.544  (se = 0.052 )
Likelihood ratio test= 2.11  on 2 df,   p=0.3
Wald test            = 1.99  on 2 df,   p=0.4
Score (logrank) test = 2.03  on 2 df,   p=0.4
```

HR = 0.9432 = e^-0.05852 = e^beta 6% lower instentaneous risk (hazard) for male (same proteins) but CI includes 1 negative beta (coefficient) is to be interpreted differently from parametric survival function (where pos coef lowers risk)

baseline of hazard (intercept) is not shown and not necesaary for HR

Including the interaction between both variables

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```
cox1 <- update(cox1, ~ . + sex:BJP)
 summary(cox1)
```

```
Call:
coxph(formula = myesurv ~ sex + BJP + sex:BJP, data = multmyel)

  n= 48, number of events= 36 

                coef exp(coef) se(coef)      z Pr(>|z|)
sexMale      -0.1357    0.8731   0.6321 -0.215    0.830
BJPNo         0.4688    1.5980   0.6380  0.735    0.463
sexMale:BJPNo 0.1131    1.1198   0.7684  0.147    0.883

              exp(coef) exp(-coef) lower .95 upper .95
sexMale          0.8731     1.1453    0.2529     3.014
BJPNo            1.5980     0.6258    0.4576     5.581
sexMale:BJPNo    1.1198     0.8930    0.2484     5.049

Concordance= 0.544  (se = 0.052 )
Likelihood ratio test= 2.14  on 3 df,   p=0.5
Wald test            = 2  on 3 df,   p=0.6
Score (logrank) test = 2.04  on 3 df,   p=0.6
```

Exercise:

Compute the model-based hazard ratios associated with the absence of the

BJ protein.

female: HR = 1.5980

male: HR = e^(0.4688+0.1131) !!!!

```
exp(0.4688+0.1131)
```

```
[1] 1.789435
```

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The Cox model with BUN, HB, age, sex, and BJP and remove interaction

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```
cox2 <- update(cox1, ~ . + BUN + HB + age - sex:BJP, multmyel)
print(summary(cox2), signif.stars = FALSE)
```

```
Call:
coxph(formula = myesurv ~ sex + BJP + BUN + HB + age, data = multmyel)

  n= 48, number of events= 36 

             coef exp(coef)  se(coef)      z Pr(>|z|)
sexMale  0.214810  1.239626  0.376100  0.571 0.567897
BJPNo    0.590635  1.805134  0.414208  1.426 0.153886
BUN      0.020706  1.020922  0.006038  3.429 0.000605
HB      -0.136414  0.872481  0.065735 -2.075 0.037968
age     -0.020520  0.979689  0.027899 -0.736 0.462035

        exp(coef) exp(-coef) lower .95 upper .95
sexMale    1.2396     0.8067    0.5931    2.5908
BJPNo      1.8051     0.5540    0.8016    4.0652
BUN        1.0209     0.9795    1.0089    1.0331
HB         0.8725     1.1462    0.7670    0.9925
age        0.9797     1.0207    0.9276    1.0348

Concordance= 0.711  (se = 0.05 )
Likelihood ratio test= 17.68  on 5 df,   p=0.003
Wald test            = 20.91  on 5 df,   p=8e-04
Score (logrank) test = 26.63  on 5 df,   p=7e-05
```

see the tests: null-hypothesis: null model is the same, covariates are not related to survival

Exercise:

Which is the hazard ratio associated with an increase of 10 units in BUN?+

coef of BUN is low because range of BUN is high -> 1 unit does not change a lot

```
exp(10*0.020706)
```

```
[1] 1.230056
```

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Model-based prediction ———————————————————–

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Estimation of the baseline hazard function

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```
coxph.detail(cox2)
```

```
$time
 [1]  1  4  5  6  8 10 12 13 14 15 16 17 18 23 24 36 40 50 51 65 66 88 91

$means
     sexMale     BJPNo       BUN        HB      age
1  0.6133566 0.7505785 100.87667  7.591568 62.99530
4  0.4186754 0.7073447  65.77149  9.481203 61.70923
5  0.5212199 0.8447387  37.12342  9.390130 63.37198
6  0.5758738 0.8258977  27.48866  9.271206 62.48314
8  0.6148257 0.8141215  28.57923  9.434710 62.68232
10 0.5650377 0.7837867  26.10754  9.548295 63.48838
12 0.5587267 0.7730509  25.74620  9.437503 63.48717
13 0.6585452 0.7325057  24.52378 10.212204 62.80387
14 0.6531423 0.7440963  24.51625 10.142775 62.75330
15 0.6331996 0.7187796  23.27399 10.113521 62.18432
16 0.6269116 0.6775208  22.93929  9.987757 62.19710
17 0.5968413 0.6515296  23.41798  9.986770 62.93837
18 0.6600478 0.6205485  23.16160 10.217381 62.91094
23 0.7536527 0.5878521  24.11872  9.972141 63.38548
24 0.7909386 0.5674616  24.32248  9.743184 63.75087
36 0.7761794 0.5369255  25.33362  9.555619 64.22746
40 0.7837057 0.5732346  24.66077  9.596441 64.05326
50 0.8618128 0.5994730  23.39311  9.879014 62.14744
51 0.8402473 0.6930263  21.26963 10.219069 60.29774
65 0.8709585 0.6612838  24.24447  9.567811 58.46963
66 0.7275532 0.2848646  20.07090 12.865974 57.88022
88 0.3962135 0.0000000  24.62272 12.188641 59.98107
91 0.0000000 0.0000000  27.00000 11.000000 58.00000

$nevent
 [1] 3 2 5 1 1 4 1 1 1 2 1 1 2 1 1 1 2 1 1 1 1 1 1

$nrisk
 [1] 48 44 42 37 35 34 28 26 25 24 21 20 19 15 14 13 12  9  8  5  4  2  1

$hazard
 [1] 0.01926605 0.02147523 0.07485200 0.01746262 0.01864378 0.08674554
 [7] 0.02532914 0.02985428 0.03032667 0.06665381 0.03821641 0.04129658
[13] 0.09299755 0.05604800 0.05882089 0.06297350 0.14357989 0.09277651
[19] 0.10725515 0.15275454 0.32251243 0.71474001 1.18376288

$score
      sexMale      BJPNo         BUN          HB         age
1   1.1599301 -0.2517355 -61.6300148 -2.97470382  10.0140885
4   0.1626492 -0.4146894  88.4570231 -2.36240573   0.5815344
5  -0.6060996 -0.2236935  21.3829178  3.74935136  10.1401197
6  -0.5758738  0.1741023 -16.4886634 -4.17120560  -1.4831411
8   0.3851743  0.1858785  24.4207654 -1.23470986  -7.6823222
10  0.7398491  0.8648532 -22.4301709  6.10682137  -6.9535308
12 -0.5587267  0.2269491 -19.7462045 -3.93750291  -3.4871663
13  0.3414548 -0.7325057   0.4762171  4.38779623   3.1961305
14  0.3468577  0.2559037  15.4837524  0.45722536   7.2467037
15 -0.2663992  0.5624408  13.4520292 -0.22704290   5.6313629
16  0.3730884  0.3224792  -5.9392861  0.01224278  -9.1971002
17 -0.5968413  0.3484704   4.5820181 -2.48677047   2.0616299
18 -0.3200956 -0.2410970 -16.3231943  1.46523821 -14.8218882
23 -0.7536527  0.4121479  -4.1187156  4.62785949  -7.3854814
24  0.2090614  0.4325384 -14.3224833  2.65681634  -6.7508677
36  0.2238206 -0.5369255  14.6663846  1.44438104  -1.2274624
40 -0.5674114 -0.1464692 -25.3215459 -3.99288250  10.8934765
50  0.1381872 -0.5994730  13.6068877 -2.17901401  11.8525619
51 -0.8402473 -0.6930263 -11.2696313 -0.11906935  -0.2977359
65  0.1290415  0.3387162   3.7555330 -2.96781091   0.5303731
66  0.2724468 -0.2848646   0.9291004 -0.06597355  -5.8802172
88  0.6037865  0.0000000  -3.6227188  1.81135942   3.0189324
91  0.0000000  0.0000000   0.0000000  0.00000000   0.0000000

$imat
, , 1

           sexMale       BJPNo         BUN          HB         age
sexMale  0.7024251   0.2049531     1.36261   -1.755045    4.017207
BJPNo    0.2049531   0.5585997   -14.43226   -1.904789    6.027067
BUN      1.3626099 -14.4322607 13320.70630 -302.589795 -125.756125
HB      -1.7550454  -1.9047889  -302.58979   34.216862  -31.420823
age      4.0172071   6.0270666  -125.75612  -31.420823  167.413589

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             sexMale        BJPNo        BUN          HB         age
sexMale   0.48602702   0.08469308  -28.29681   0.4656919    1.370607
BJPNo     0.08469308   0.41085227  -29.24835  -0.5822700    3.756980
BUN     -28.29681284 -29.24834738 7266.34979  37.7345600 -342.372284
HB        0.46569188  -0.58227004   37.73456  11.6684927  -10.084365
age       1.37060726   3.75698016 -342.37228 -10.0843650  135.297501

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            sexMale      BJPNo         BUN         HB         age
sexMale   1.2451402 -0.1480635  -15.798422   2.512404  -2.7090467
BJPNo    -0.1480635  0.6554661    4.766843  -1.293705  -0.2006521
BUN     -15.7984219  4.7668428 5523.226634  11.813968 442.3959089
HB        2.5124041 -1.2937055   11.813968  37.937420 -19.2744906
age      -2.7090467 -0.2006521  442.395909 -19.274491 218.3850650

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            sexMale       BJPNo         BUN         HB        age
sexMale  0.24424317 -0.02888873   2.3733307  0.6543219 -0.2884268
BJPNo   -0.02888873  0.14379070  -0.6833734 -0.3079821 -0.2708169
BUN      2.37333070 -0.68337339 216.0775289 -6.2067875 27.0269316
HB       0.65432188 -0.30798211  -6.2067875  8.7185897 -4.9945293
age     -0.28842681 -0.27081688  27.0269316 -4.9945293 41.2906961

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            sexMale       BJPNo         BUN         HB        age
sexMale  0.23681506 -0.02360244   1.8633506  0.5980533 -0.4303976
BJPNo   -0.02360244  0.15132770  -0.5268827 -0.2984220 -0.2521113
BUN      1.86335065 -0.52688271 211.9043310 -9.5164330 25.4890895
HB       0.59805334 -0.29842197  -9.5164330  8.5249715 -5.5452602
age     -0.43039757 -0.25211133  25.4890895 -5.5452602 43.2064213

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           sexMale      BJPNo        BUN         HB         age
sexMale  0.9828572 -0.1531280   3.504027   2.988544  -1.2107459
BJPNo   -0.1531280  0.6775769  -4.587469  -1.290636  -0.4756841
BUN      3.5040269 -4.5874687 669.359045 -24.975136 192.4382220
HB       2.9885436 -1.2906360 -24.975136  37.462651 -28.9087883
age     -1.2107459 -0.4756841 192.438222 -28.908788 159.9371643

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            sexMale       BJPNo         BUN         HB        age
sexMale  0.24655117 -0.03494377   0.4734846  0.7933314 -0.5509225
BJPNo   -0.03494377  0.17544318  -0.8875180 -0.2880057 -0.2284764
BUN      0.47348457 -0.88751800 170.8890871 -7.5473377 50.3623363
HB       0.79333141 -0.28800571  -7.5473377  9.7147088 -7.2786060
age     -0.55092252 -0.22847642  50.3623363 -7.2786060 40.2772754

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            sexMale       BJPNo        BUN         HB        age
sexMale  0.22486342 -0.01448572   1.363094  0.4248873 -0.1993649
BJPNo   -0.01448572  0.19594110  -1.373067 -0.1322309 -0.4520725
BUN      1.36309412 -1.37306699 127.854327 -1.6922676 36.3298989
HB       0.42488730 -0.13223087  -1.692268  7.4764198 -4.8238142
age     -0.19936492 -0.45207253  36.329899 -4.8238142 39.5748773

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            sexMale       BJPNo        BUN          HB        age
sexMale  0.22654744 -0.01069464   1.382049  0.40752836 -0.2200612
BJPNo   -0.01069464  0.19041699  -1.389186 -0.08266123 -0.4215945
BUN      1.38204901 -1.38918637 129.873751 -1.75263129 36.8802906
HB       0.40752836 -0.08266123  -1.752631  7.28525972 -5.1255584
age     -0.22006120 -0.42159450  36.880291 -5.12555839 40.0368839

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           sexMale      BJPNo        BUN          HB         age
sexMale  0.4644982 -0.0346720   2.026457   0.8356278  -0.9221836
BJPNo   -0.0346720  0.4041293  -3.755637  -0.1981153  -1.2481446
BUN      2.0264573 -3.7556369 242.348786  -5.5286999  61.7461072
HB       0.8356278 -0.1981153  -5.528700  15.8931392 -11.9614367
age     -0.9221836 -1.2481446  61.746107 -11.9614367  79.2579528

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            sexMale       BJPNo         BUN         HB        age
sexMale  0.23389345 -0.02193578   0.9281892  0.3697677 -0.5330387
BJPNo   -0.02193578  0.21848635  -2.2591320 -0.1541561 -0.7106375
BUN      0.92818917 -2.25913198 121.3273986 -3.0893920 27.4477574
HB       0.36976775 -0.15415613  -3.0893920  8.5780061 -6.2859842
age     -0.53303868 -0.71063746  27.4477574 -6.2859842 41.3009721

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            sexMale       BJPNo        BUN         HB        age
sexMale  0.24062176 -0.03418236   1.195990  0.3991726 -0.2771513
BJPNo   -0.03418236  0.22703878  -2.274403 -0.1669247 -0.5096030
BUN      1.19598995 -2.27440264 128.033909 -3.3320586 24.9025410
HB       0.39917256 -0.16692469  -3.332059  9.2693651 -6.7828166
age     -0.27715128 -0.50960296  24.902541 -6.7828166 37.2627424

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            sexMale       BJPNo        BUN          HB         age
sexMale  0.44873164 -0.04445649   3.303596   0.4888853  -0.3034444
BJPNo   -0.04445649  0.47091318  -5.457680  -0.2650344  -1.2455071
BUN      3.30359570 -5.45768017 279.991380  -4.2170589  50.4097481
HB       0.48888534 -0.26503438  -4.217059  18.8908179 -13.1468575
age     -0.30344444 -1.24550706  50.409748 -13.1468575  79.3534652

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            sexMale       BJPNo         BUN         HB         age
sexMale  0.18566030 -0.01805007   1.7388915  0.3039484 -0.03998068
BJPNo   -0.01805007  0.24228202  -3.1274335 -0.3837728 -0.68600889
BUN      1.73889154 -3.12743348 160.8929883 -0.5698879 24.05105950
HB       0.30394841 -0.38377278  -0.5698879  9.8006037 -5.68515732
age     -0.03998068 -0.68600889  24.0510595 -5.6851573 38.14695470

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             sexMale        BJPNo         BUN         HB        age
sexMale  0.165354721 -0.002815485   1.6637530  0.5000766 -0.3309568
BJPNo   -0.002815485  0.245448930  -3.1940215 -0.5017920 -0.5619046
BUN      1.663753040 -3.194021519 167.9721608  0.3915799 23.6615756
HB       0.500076649 -0.501792025   0.3915799  9.1734734 -4.1918082
age     -0.330956831 -0.561904605  23.6615756 -4.1918082 37.2021606

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            sexMale       BJPNo        BUN         HB        age
sexMale  0.17372494 -0.00984887   2.007522  0.4934000 -0.2476499
BJPNo   -0.00984887  0.24863651  -2.951282 -0.6240737 -0.3808748
BUN      2.00752211 -2.95128206 164.326272  3.2952695 18.0241054
HB       0.49339999 -0.62407374   3.295269  9.2875926 -3.1321227
age     -0.24764987 -0.38087484  18.024105 -3.1321227 36.3839717

, , 40

            sexMale       BJPNo         BUN         HB        age
sexMale  0.33762460  0.02972164   3.3752761  0.7799260 -0.1766467
BJPNo    0.02972164  0.48910958  -5.0664091 -1.1126669 -1.1223720
BUN      3.37527614 -5.06640914 319.1293718 -0.1478829 51.2631613
HB       0.77992602 -1.11266692  -0.1478829 19.1255619 -4.7535831
age     -0.17664673 -1.12237195  51.2631613 -4.7535831 79.3210123

, , 50

            sexMale       BJPNo         BUN          HB        age
sexMale  0.11909153  0.08283952   0.5183953  -0.1010743  0.4534970
BJPNo    0.08283952  0.24010513  -1.0136753  -0.2406200 -0.4199935
BUN      0.51839532 -1.01367528  68.3173360 -17.1421474 15.4446986
HB      -0.10107426 -0.24062002 -17.1421474   8.2211138 -1.5640810
age      0.45349704 -0.41999355  15.4446986  -1.5640810 34.9239360

, , 51

            sexMale      BJPNo         BUN           HB         age
sexMale  0.13423174  0.1107128   0.2600640  -0.06252309   0.2287746
BJPNo    0.11071278  0.2127409   0.2997594  -0.51383829   0.7963548
BUN      0.26006400  0.2997594  45.5757506 -14.46814254 -11.2415105
HB      -0.06252309 -0.5138383 -14.4681425   8.64747185   2.8513578
age      0.22877463  0.7963548 -11.2415105   2.85135777  15.0290321

, , 65

            sexMale       BJPNo         BUN          HB         age
sexMale  0.11238979  0.08533306  -0.3555781  -0.1848118  0.06060137
BJPNo    0.08533306  0.22398753   0.3247044  -0.9641348  0.48565017
BUN     -0.35557813  0.32470445  30.8393728 -15.2081437 -0.26777859
HB      -0.18481183 -0.96413476 -15.2081437  10.5205829 -0.91362481
age      0.06060137  0.48565017  -0.2677786  -0.9136248  7.46232387

, , 66

            sexMale       BJPNo       BUN         HB         age
sexMale  0.19821955  0.07761045 -1.887811  0.5083786 -0.03263444
BJPNo    0.07761045  0.20371676 -2.299114  0.3230440  0.60385106
BUN     -1.88781138 -2.29911353 32.018890 -5.9576672 -5.23885293
HB       0.50837856  0.32304398 -5.957667  1.5460286  1.76427909
age     -0.03263444  0.60385106 -5.238853  1.7642791 15.09527998

, , 88

           sexMale BJPNo       BUN         HB       age
sexMale  0.2392284     0 -1.435370  0.7176851  1.196142
BJPNo    0.0000000     0  0.000000  0.0000000  0.000000
BUN     -1.4353702     0  8.612221 -4.3061106 -7.176851
HB       0.7176851     0 -4.306111  2.1530553  3.588426
age      1.1961418     0 -7.176851  3.5884255  5.980709

, , 91

        sexMale BJPNo           BUN            HB           age
sexMale       0     0  0.000000e+00  0.000000e+00  0.000000e+00
BJPNo         0     0  0.000000e+00  0.000000e+00  0.000000e+00
BUN           0     0  4.061305e-15 -4.277703e-16  2.102785e-15
HB            0     0 -4.277703e-16  3.160938e-17 -4.107003e-17
age           0     0  2.102785e-15 -4.107003e-17 -6.571204e-16


$varhaz
 [1] 0.0001262169 0.0002317520 0.0011259881 0.0003049429 0.0003475905
 [6] 0.0018840498 0.0006415654 0.0008912780 0.0009197070 0.0022233553
[11] 0.0014604938 0.0017054076 0.0043268232 0.0031413781 0.0034598976
[16] 0.0039656623 0.0103320428 0.0086074812 0.0115036666 0.0233339500
[21] 0.1040142696 0.5108532873 1.4012945552

$wtrisk
 [1] 186.0465357 100.2374860  73.6918331  57.2651896  53.6371909  48.5910406
 [7]  39.4802179  33.4960347  32.9742750  30.9316073  26.1667718  24.2150796
[13]  22.0413294  17.8418504  17.0007616  15.8796942  14.6426881  10.7785902
[19]   9.3235619   6.5464502   3.1006557   1.3991101   0.8447638

$y
      time status
1  -1   13      1
2  -1   52      0
3  -1    5      1
4  -1   40      1
5  -1   10      1
6  -1    7      0
7  -1   66      1
8  -1   10      0
9  -1   10      1
10 -1   14      1
11 -1   15      1
12 -1    4      1
13 -1   65      1
14 -1    5      1
15 -1   11      0
16 -1   10      1
17 -1   15      0
18 -1    5      1
19 -1   76      0
20 -1   56      0
21 -1   88      1
22 -1   24      1
23 -1   51      1
24 -1    4      1
25 -1   40      0
26 -1    8      1
27 -1   18      1
28 -1    5      1
29 -1   16      1
30 -1   50      1
```

```
31 -1   40      1
32 -1    1      1
33 -1   36      1
34 -1    5      1
35 -1   10      1
36 -1   91      1
37 -1   18      0
38 -1    1      1
39 -1   18      0
40 -1    6      1
41 -1    1      1
42 -1   23      1
43 -1   15      1
44 -1   18      1
45 -1   12      0
46 -1   12      1
47 -1   17      1
48 -1    3      0

$x
   sexMale BJPNo BUN   HB age
1        1     0  25 14.6  66
2        1     1  13 12.0  66
3        0     0  15 11.4  53
4        1     0  10 10.2  69
5        1     1  20 13.2  65
6        0     1  12  9.9  57
7        1     0  21 12.8  52
8        1     0  47 14.0  60
9        1     1  37  7.5  70
10       1     1  40 10.6  70
11       1     1  39 11.2  68
12       0     0 172 10.1  50
13       1     1  28  6.6  59
14       1     1  13  9.7  60
15       0     1  25  8.8  66
16       0     1  12  9.6  51
17       1     1  14 13.0  55
18       0     1  26 10.4  67
19       1     1  12 14.0  60
20       1     1  18 12.5  66
21       1     0  21 14.0  63
22       1     1  10 12.4  57
23       0     0  10 10.1  60
24       1     1  48  6.5  74
25       1     0  57 12.8  72
26       1     1  53  8.2  55
27       1     1  12 14.4  51
28       0     1 130 10.2  70
29       1     1  17 10.0  53
30       1     0  37  7.7  74
31       0     1  14  5.0  70
32       1     1 165  3.4  67
33       1     0  40 11.0  63
34       1     1  23  9.0  77
35       1     1  13 14.0  61
36       0     0  27 11.0  58
37       0     1  21 10.8  69
38       1     0  20  5.1  57
39       0     1  21 13.0  59
40       0     1  11  5.1  61
41       1     1  56 11.3  75
42       0     1  20 14.6  56
43       0     1  21  8.8  62
44       0     0  18  7.5  60
45       0     1  46  4.9  71
46       0     1   6  5.5  60
47       0     1  28  7.5  65
48       1     0  90 10.2  59
attr(,"assign")
[1] 1 2 3 4 5
attr(,"contrasts")
attr(,"contrasts")$sex
[1] "contr.treatment"

attr(,"contrasts")$BJP
[1] "contr.treatment"
```

```
coxph.detail(cox2)$hazard
```

```
 [1] 0.01926605 0.02147523 0.07485200 0.01746262 0.01864378 0.08674554
 [7] 0.02532914 0.02985428 0.03032667 0.06665381 0.03821641 0.04129658
[13] 0.09299755 0.05604800 0.05882089 0.06297350 0.14357989 0.09277651
[19] 0.10725515 0.15275454 0.32251243 0.71474001 1.18376288
```

(Predicted) survival function of an "average" patient

```
cox2$mean
```

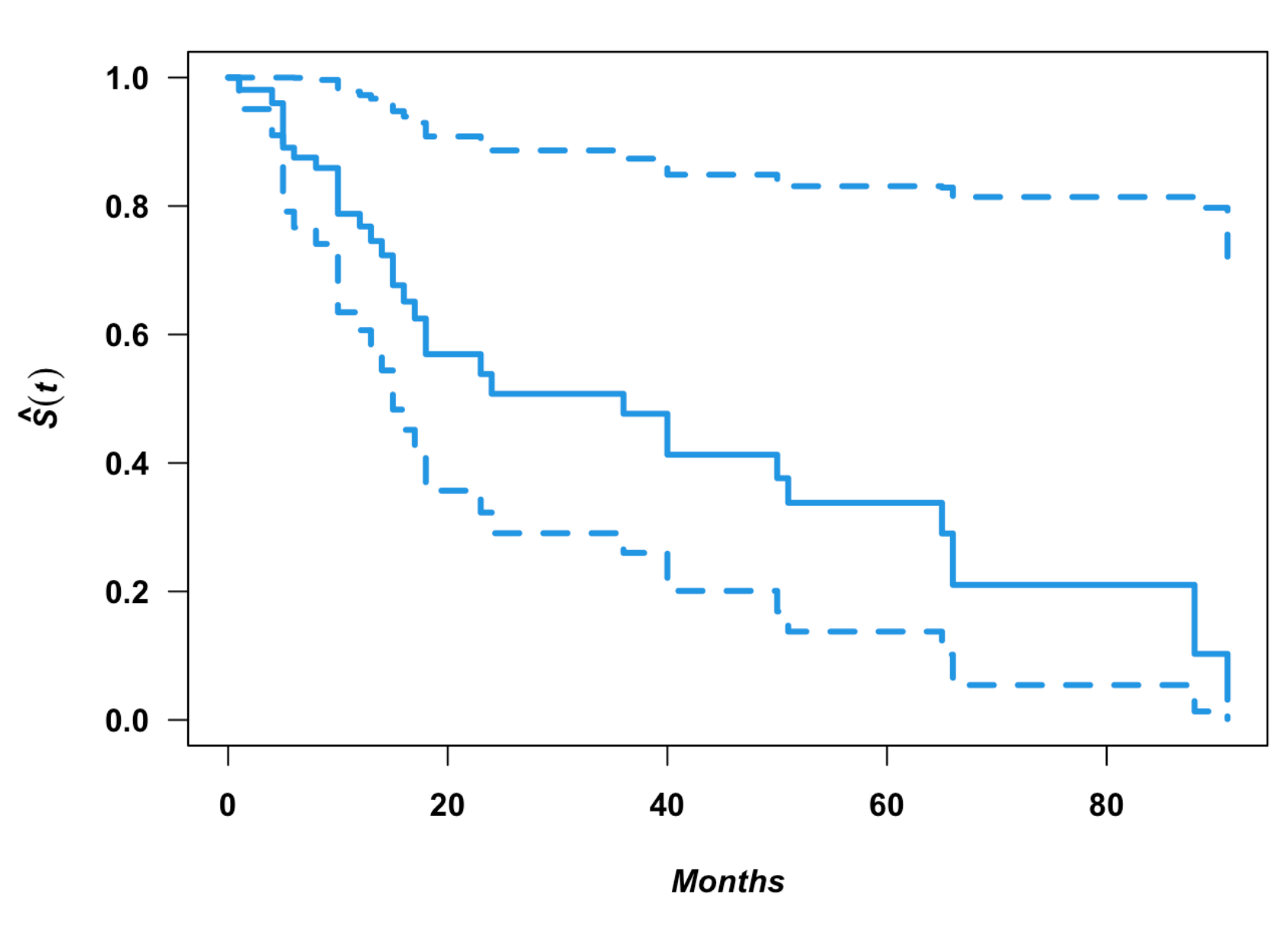
```
 sexMale    BJPNo      BUN       HB      age 
 0.00000  0.00000 34.04167 10.12708 62.68750 
```

```
survfit(cox2)
```

```
Call: survfit(formula = cox2)

      n events median 0.95LCL 0.95UCL
[1,] 48     36     36      15      NA
```

```
#windows(width = 8)
par(font = 2, font.lab = 4, font.axis = 2, las = 1, mar = c(5, 5, 2, 2))
plot(survfit(cox2), lwd = 3, col = 4, xlab = "Months", ylab = expression(bolditalic(hat(
```



Patient profiles for prediction

——————————-

```
#create dataframe of 5 columns (each variable)
(newdat <- data.frame(BUN = 25, sex = factor(rep(c("Female", "Male"), 2)), HB = 10, age
```

```
  BUN    sex HB age BJP
1  25 Female 10  65  No
2  25   Male 10  65  No
3  25 Female 10  65 Yes
4  25   Male 10  65 Yes
```

```
survfit(cox2, newdata = newdat)
```

```
Call: survfit(formula = cox2, newdata = newdat)

   n events median 0.95LCL 0.95UCL
1 48     36     18      13      NA
2 48     36     15      10      40
3 48     36     40      17      NA
4 48     36     36      15      NA
```

```
summary(survfit(cox2, newdata = newdat))
```

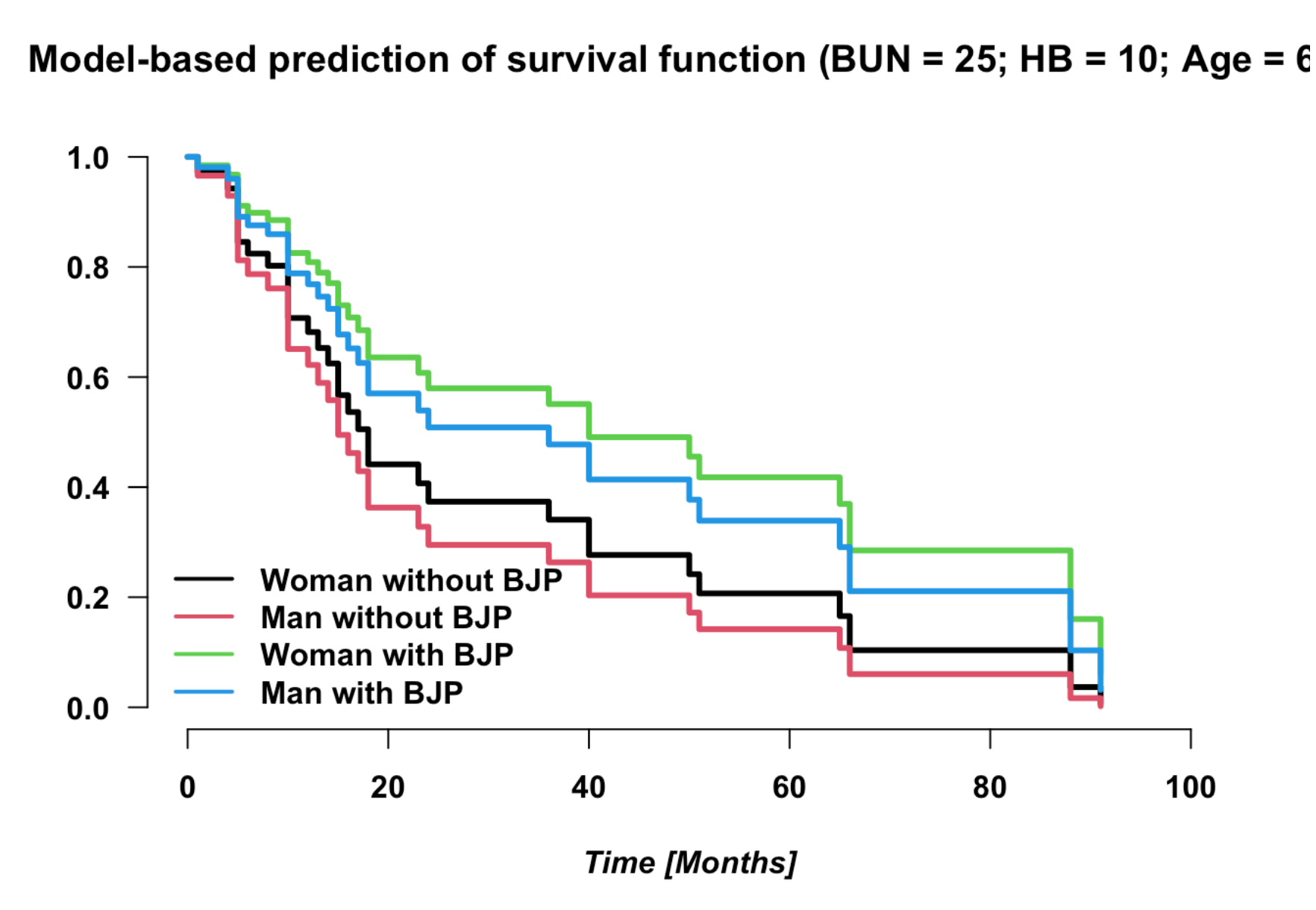
```
Call: survfit(formula = cox2, newdata = newdat)

 time n.risk n.event survival1 survival2 survival3 survival4
    1     48       3   0.97240   0.96590    0.9846    0.9810
    4     44       2   0.94254   0.92927    0.9677    0.9602
    5     42       5   0.84544   0.81210    0.9112    0.8911
    6     37       1   0.82426   0.78696    0.8985    0.8757
    8     35       1   0.80224   0.76098    0.8851    0.8596
   10     34       4   0.70727   0.65094    0.8254    0.7883
   12     28       1   0.68172   0.62192    0.8088    0.7687
   13     26       1   0.65279   0.58937    0.7896    0.7461
   14     25       1   0.62466   0.55805    0.7705    0.7239
   15     24       2   0.56702   0.49494    0.7303    0.6773
   16     21       1   0.53640   0.46203    0.7082    0.6520
   17     20       1   0.50517   0.42892    0.6850    0.6257
   18     19       2   0.44134   0.36279    0.6356    0.5702
   23     15       1   0.40683   0.32796    0.6076    0.5392
   24     14       1   0.37352   0.29500    0.5795    0.5085
   36     13       1   0.34087   0.26338    0.5509    0.4775
   40     12       2   0.27670   0.20338    0.4908    0.4138
   50      9       1   0.24182   0.17209    0.4555    0.3772
   51      8       1   0.20693   0.14187    0.4178    0.3390
   65      5       1   0.16575   0.10775    0.3695    0.2911
   66      4       1   0.10376   0.06029    0.2850    0.2110
   88      2       1   0.03674   0.01665    0.1604    0.1034
   91      1       1   0.00658   0.00198    0.0619    0.0318
```

Graphical respresentation

———————-

```
#windows(width = 8)
par(font = 2, font.lab = 4, font.axis = 2, las = 1)
plot(survfit(cox2, newdata = newdat), col = 1:4, lwd = 3, xlab = "Time [Months]",
     bty = "n", xlim = c(0, 100))
legend("bottomleft", c("Woman without BJP", "Man without BJP",
                       "Woman with BJP", "Man with BJP"),
       col = 1:4, lwd = 2, bty = "n")
title("Model-based prediction of survival function (BUN = 25; HB = 10; Age = 65)")
```



use AFT model to predict a survival probability/time (parametric)

Residuals and goodness-of-fit of the Cox model

——————————————————————-

```
residuals(cox2)
```

```
          1           2           3           4           5           6 
 0.84679641 -1.13595691  0.94180253  0.44492536  0.78493770 -0.17638830 
          7           8           9          10          11          12 
-0.27638401 -0.24083066  0.39942085  0.33826696  0.28268345  0.20706275 
         13          14          15          16          17          18 
-3.26357857  0.86352516 -0.39966115  0.67983589 -0.45668977  0.88654760 
         19          20          21          22          23          24 
-1.37775573 -1.17680056 -0.26090548  0.23992916  0.30077546  0.86405652 
         25          26          27          28          29          30 
-1.01186168  0.23450062  0.45745778  0.05559857  0.16306455 -0.42202601 
         31          32          33          34          35          36 
-0.74862485  0.06570904  0.08342539  0.86969761  0.81892128 -1.92148329 
         37          38          39          40          41          42 
-0.62188667  0.97497068 -0.56557546  0.69366168  0.97175306  0.47922500 
         43          44          45          46          47          48 
 0.37452761  0.41498475 -1.04923626  0.47076216 -0.02192804 -0.08125222 
```

The Schoenfeld residuals:

Can we assume that the assumption of proportional hazards holds?

————————————————————————

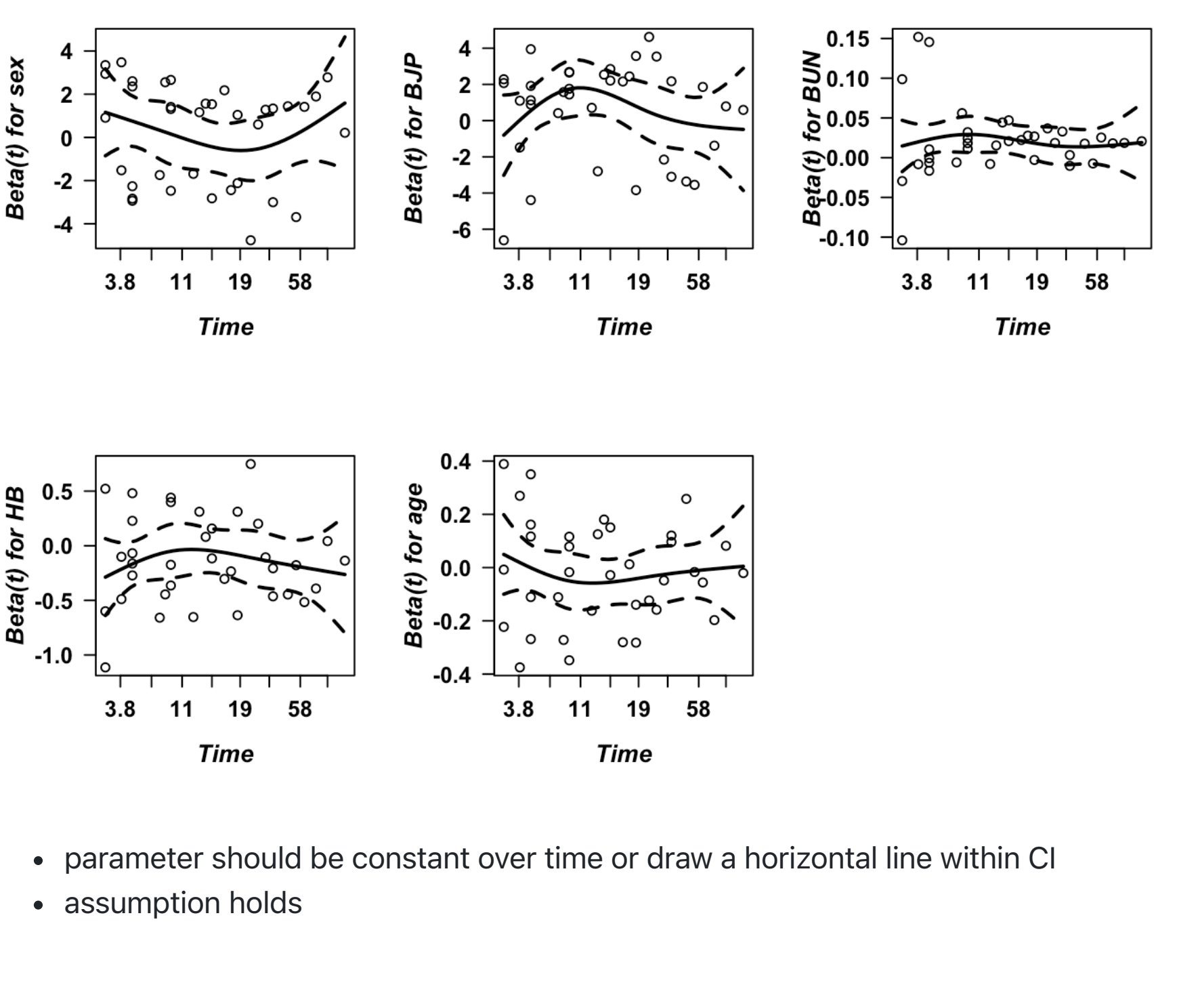
```
residuals(cox2, "schoenfeld") ## (i) Use of function cox.zph cox.zph(cox2)
```

```
      sexMale      BJPNo         BUN          HB         age
1   0.3866434  0.2494215  64.1233284 -4.19156794   4.0046962
1   0.3866434 -0.7505785 -80.8766716 -2.49156794  -5.9953038
1   0.3866434  0.2494215 -44.8766716  3.70843206  12.0046962
4  -0.4186754 -0.7073447 106.2285115  0.61879713 -11.7092328
4   0.5813246  0.2926553 -17.7714885 -2.98120287  12.2907672
5  -0.5212199 -0.8447387 -22.1234164  2.00987027 -10.3719761
5   0.4787801  0.1552613 -24.1234164  0.30987027  -3.3719761
5  -0.5212199  0.1552613 -11.1234164  1.00987027   3.6280239
5  -0.5212199  0.1552613  92.8765836  0.80987027   6.6280239
5   0.4787801  0.1552613 -14.1234164 -0.39012973  13.6280239
6  -0.5758738  0.1741023 -16.4886634 -4.17120560  -1.4831411
8   0.3851743  0.1858785  24.4207654 -7.23470986  -7.6823222
10  0.4349623  0.2162133  -6.1075427  3.65170534   1.5116173
10  0.4349623  0.2162133  10.8924573 -2.04829466   6.5116173
10 -0.5650377  0.2162133 -14.1075427  0.05170534 -12.4883827
10  0.4349623  0.2162133 -13.1075427  4.45170534  -2.4883827
12 -0.5587267  0.2269491 -19.7462045 -3.93750291  -3.4871663
13  0.3414548 -0.7325057   0.4762171  4.38779623   3.1961305
14  0.3468577  0.2559037  15.4837524  0.45722536   7.2467037
15  0.3668004  0.2812204  15.7260146  1.08647855   5.8156815
15 -0.6331996  0.2812204  -2.2739854 -1.31352145  -0.1843185
16  0.3730884  0.3224792  -5.9392861  0.01224278  -9.1971002
17 -0.5968413  0.3484704   4.5820181 -2.48677047   2.0616299
18  0.3399522  0.3794515 -11.1615972  4.18261911 -11.9109441
18 -0.6600478 -0.6205485  -5.1615972 -2.71738089  -2.9109441
23 -0.7536527  0.4121479  -4.1187156  4.62785949  -7.3854814
24  0.2090614  0.4325384 -14.3224833  2.65681634  -6.7508677
36  0.2238206 -0.5369255  14.6663846  1.44438104  -1.2274624
40  0.2162943 -0.5732346 -14.6607729  0.60355875   4.9467382
40 -0.7837057  0.4267654 -10.6607729 -4.59644125   5.9467382
50  0.1381872 -0.5994730  13.6068877 -2.17901401  11.8525619
51 -0.8402473 -0.6930263 -11.2696313 -0.11906935  -0.2977359
65  0.1290415  0.3387162   3.7555330 -2.96781091   0.5303731
66  0.2724468 -0.2848646   0.9291004 -0.06597355  -5.8802172
88  0.6037865  0.0000000  -3.6227188  1.81135942   3.0189324
91  0.0000000  0.0000000   0.0000000  0.00000000   0.0000000
```

- schoenfeld residuals only exists with censored events
- prportion holds for spec covariate = no correlation between survival time and residuals of a specific covariate

(ii) Use of function plot.cox.zph

```
#windows(width = 12, height = 7)
par(mfrow = c(2, 3), font = 2, font.lab = 4, font.axis = 2, las = 1, cex.lab = 1.3, cex.
plot(cox.zph(cox2), lwd = 2)
```



- parameter should be constant over time or draw a horizontal line within CI
- assumption holds